



INSTALLATION INSTRUCTIONS

Installation and Setup

Install, pair, and play with the VirtualGlove Controller and supported consoles.

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This guide takes you from a prepared Arduino UNO Q and game system to your first working VirtualGlove game. You will install VirtualGlove on both devices, pair them securely, center your hand, and test the controls.

VirtualGlove 0.5.0 supports RetroPie, Recalbox, Batocera, and LaunchBox on Windows. The normal commands below install the latest stable release. If you need a specific version or a development build, use the [technical installation reference](#).

1. Before you begin

You need:

- a provisioned **VirtualGlove Controller** running on an Arduino UNO Q;
- a UVC USB camera connected through a powered USB hub;
- a supported game system: RetroPie, Recalbox, Batocera, or LaunchBox;
- a physical controller already working with that game system;
- both devices on the same trusted local network with internet access; and
- your own legally obtained NES games. VirtualGlove includes no ROMs or BIOS files.

Close RetroArch and any running game before installation. Keep both devices powered and connected while the installers run. First installation can take several minutes.

Choose your console section

Platform	Where you install	How Player 1 works
RetroPie	Raspberry Pi terminal	VirtualGlove appears beside your physical controller.
Recalbox 10.x	Recalbox terminal as root	Your selected physical controller and VirtualGlove share VirtualGlove Merged Player 1 .
Batocera 38 or newer	Batocera terminal as root	Your selected physical controller and VirtualGlove share VirtualGlove Merged Player 1 .
LaunchBox	Windows PowerShell	Your physical XInput controller remains available while VirtualGlove supplies a managed RetroArch controller.

Most NES games use the included FCEUmm core. Super Glove Ball also supports the special **Nestopia (VirtualGlove)** core, which provides its native three-axis movement and glove actions. FCEUmm remains a complete joystick-mode fallback.

Open a Controller terminal

For a new UNO Q, first complete Arduino App Lab setup and connect it to your network. App Lab shows its current name or IP address. From another computer, open a terminal and connect with:

```
SH
ssh arduino@UNO-Q-NAME.local
```

Use the password you created during board setup. The installer may request it again when administrator access is needed, but it does not save the password.

2. Install the VirtualGlove Controller

In the UNO Q terminal, copy and run these two lines:

```
SH
cd /home/arduino
curl -fLO \
  https://github.com/mathan416/VirtualGlove/releases/latest/download/install-uno-q.sh \
  && bash install-uno-q.sh
```

The installer verifies the release, installs VirtualGlove, loads the matching Matrix firmware, and configures automatic startup. It also installs the guarded camera-recovery and shutdown helpers used by the Controller.

Questions the installer asks

On a first installation, the installer shows the UNO Q's current name and suggests **virtualglove**. Press Enter to use [virtualglove.local](#), or enter a different short name such as [games-room](#). Confirm the displayed address before continuing. Updates keep the established name and do not ask again.

The camera can be connected before or after installation. Camera recovery is configured automatically when a supported camera is available.

What is preserved

Rerunning the installer updates VirtualGlove without erasing players, calibration, gesture tuning, dead-zone settings, camera choices, pairing, or device configuration. The installer prints the location of any backup it creates.

Controller checkpoint

Open the Dashboard address printed by the installer, normally:

<http://virtualglove.local:8088/dashboard>

The Dashboard should load. With gestures off, a closed camera is normal. Open **Play** or **Glove Academy** to confirm that the camera can show your whole hand, then return to Dashboard with controller output stopped.

For the simplest first setup, leave **Camera**, frame rate, exposure, and camera reader on their recommended automatic settings. If tracking is delayed or unstable, use **Find the best camera settings** in Setup. The [Camera Guide](#) explains the advanced choices.

Camera

Camera

Camera frame rate

Camera buffers

Camera reader

Automatic — choose the connected

30 fps

1 buffer — lowest queue depth

Recommended — OpenCV

Exposure behavior

Hand or glove (diagnostic label)

Automatic — fixed frame rate

Bare hand

Camera is delivering 30 fps. Reader: ~; automatic fixed-rate exposure unavailable.

- Automatic and OpenCV are the recommended starting point and work when the camera is connected later.
- Direct V4L2 remains available for engineering comparisons and falls back safely when unsupported.
- Manual exposure is optional, camera-dependent, and restored to automatic when tracking closes.

[Open the Camera guide](#) for compatibility, exposure, lighting, and troubleshooting.

Save camera settings

Find the best camera settings

Pixel Pal can compare safe settings for this camera. Keep one hand visible and follow the short movement cues. No video or images are saved.

Start camera test

Camera test cancelled

Your original camera settings are restored. You can start a new test.

Pixel Pal recommends OpenCV recommended. Your previous settings are still active until you choose Use recommended settings.

Advanced camera settings with Automatic, a discovered camera, and optional manual exposure

3. Install the console

Choose only the section for your game system. Close RetroArch first. On a new installation, the console installer asks for the Controller name or IP address; enter the name selected above, normally [virtualglove.local](#).

Each section uses the same latest-release installer as future updates. The installer preserves ROMs, saves, pairing, game registrations, and unrelated controller settings.

RetroPie

Before you start

Confirm that a physical controller works in EmulationStation and that RetroPie has internet access. Connect to RetroPie as its normal user, without starting a root shell.

Install

Run this command in the RetroPie terminal:

```
SH
curl -fLO \
  https://github.com/mathan416/VirtualGlove/releases/latest/download/install-retropie.sh \
  && bash install-retropie.sh
```

Questions the installer asks

- Enter the Controller name or address on a first installation.
- If a required emulator is missing, the installer offers to add it through RetroPie Setup.
- If Super Glove Ball is registered, the installer can build the optional [lr-nestopia-powerglove](#) core. Declining leaves FCEUmm available.
- The optional **VirtualGlove Calibration Test** can be added to the Ports menu now or on a later installer run.

If an older Buster-based RetroPie reports that its Raspbian repository has no Release file, stop and follow [Buster package source moved](#), then run the installer again.

What the installer configures

VirtualGlove appears as a separate game controller. Your physical controller remains available for menus and gameplay. Registered games tell VirtualGlove which profile to use, and game exit safely releases all controls.

For Super Glove Ball, FCEUmm is the safe fallback. To use native glove control, choose [lr-nestopia-powerglove](#) for that ROM from RetroPie's launch menu.

Checkpoint

The final report should confirm that the receiver starts automatically. It is normal to see **ACTION** for pairing or physical gameplay checks that still need you.

First game and updates

After pairing, test an ordinary registered game with FCEUmm and confirm both VirtualGlove and the physical controller work. Test Super Glove Ball separately if you installed its native core. To update later, close the game and rerun the same install command.

Recalbox 10.x

Before you start

Confirm that your intended physical Player 1 controller works in EmulationStation. Connect to Recalbox over SSH as `root`; do not add `sudo` to the commands below.

Install

Run this command in the Recalbox terminal:

```
SH
curl -fLO \
  https://github.com/mathan416/VirtualGlove/releases/latest/download/install-recalbox.sh \
  && bash install-recalbox.sh
```

Questions the installer asks

- Enter the Controller name or address on a first installation.
- If exactly one configured controller is available, it is selected as physical Player 1 automatically.
- If several controllers are available, choose the one you want VirtualGlove to share with. The installer displays the available choices.

What the installer configures

Recalbox receives one game controller named **VirtualGlove Merged Player 1**. It combines the selected physical controller with VirtualGlove during NES gameplay. The original controller continues to operate EmulationStation without duplicate menu movement. Player 2 and unrelated settings are preserved.

Registered Super Glove Ball filenames use the packaged Nestopia (VirtualGlove) core when a compatible core is available and you have not already made an explicit per-game choice. FCEUmm remains available if the native core cannot be used.

Checkpoint

The final report should confirm automatic startup, the selected physical Player 1 controller, and creation of **VirtualGlove Merged Player 1**. Pairing and live gameplay may still appear as `ACTION` items.

First game and updates

After pairing, test an ordinary registered game with FCEUmm. Confirm that hand controls and the physical controller both operate Player 1. Then test Super Glove Ball if it is installed. To update later, close the game and rerun the same install command.

Batocera 38 and newer

Before you start

Confirm that your intended physical Player 1 controller works in EmulationStation. Connect to Batocera over SSH as `root`; do not add `sudo` to the commands below.

Install

Run this command in the Batocera terminal:

```
SH
curl -fLO \
  https://github.com/mathan416/VirtualGlove/releases/latest/download/install-batocera.sh \
  && bash install-batocera.sh
```

Questions the installer asks

- Enter the Controller name or address on a first installation.
- If exactly one configured controller is available, it is selected as physical Player 1 automatically.
- If several controllers are available, choose the one you want VirtualGlove to share with. The installer displays the available choices.

What the installer configures

Batocera receives one game controller named **VirtualGlove Merged Player 1**. It combines the selected physical controller with VirtualGlove during NES gameplay. The original controller continues to operate EmulationStation without duplicate menu movement. Player 2 and unrelated settings are preserved.

Registered Super Glove Ball filenames use the packaged Nestopia (VirtualGlove) core when a compatible core is available and you have not already made an explicit per-game choice. FCEUmm remains available if the native core cannot be used.

Checkpoint

The final report should confirm automatic startup, the selected physical Player 1 controller, and creation of **VirtualGlove Merged Player 1**. Pairing and live gameplay may still appear as `ACTION` items.

First game and updates

After pairing, test an ordinary registered game with FCEUmm. Confirm that hand controls and the physical controller both operate Player 1. Then test Super Glove Ball if it is installed. To update later, close the game and rerun the same install command.

LaunchBox on Windows x86-64

Before you start

Install 64-bit Python with its [py](#) launcher and 64-bit RetroArch with FCEUmm. Confirm that your physical XInput controller works in RetroArch. Close LaunchBox, Big Box, and RetroArch before installation.

Download [VirtualGlove-LaunchBox.zip](#) from the latest VirtualGlove release and extract it to a temporary folder. Open PowerShell as Administrator, using the same Windows account that runs LaunchBox.

Install

From the extracted package folder, run:

```
POWERSHELL
powershell -ExecutionPolicy Bypass -File .\launchbox\install-launchbox.ps1 `
  -LaunchBoxRoot "C:\LaunchBox" -RetroArchRoot "C:\RetroArch" `
  -ControllerHost "virtualglove.local"
```

Change the two folders and Controller name to match your system.

Questions the installer asks

- Confirm the LaunchBox and RetroArch folders if your installation uses different locations.
- Allow the authenticated VirtualGlove receiver on Private networks if Windows asks. Do not enable it for Public networks.

What the installer configures

LaunchBox gains a **VirtualGlove RetroArch** emulator for NES games. Ordinary games use FCEUmm with VirtualGlove's managed RetroArch controller. Your physical XInput controller and real keyboard remain available. Super Glove Ball uses Nestopia (VirtualGlove) when its exact filename is registered.

Existing NES games assigned to standard RetroArch are moved to VirtualGlove RetroArch. Games explicitly assigned to another emulator remain unchanged. ROMs, saves, pairing, and unrelated LaunchBox settings are preserved.

Checkpoint

The installer should confirm the VirtualGlove RetroArch entry and its managed receiver. LaunchBox does not have a separate check-only command, so the first registered game is the final test.

First game and updates

After pairing, launch an ordinary registered game from LaunchBox. Confirm that VirtualGlove directions and buttons work while the physical XInput controller still controls Player 1. Test Super Glove Ball separately. To update later, close LaunchBox, Big Box, and RetroArch, extract the new package, and rerun the same installer.

Add NES ROMs after VirtualGlove is installed

Adding a ROM does not require reinstalling VirtualGlove:

1. Copy or import the [.nes](#), [.zip](#), or [.7z](#) game and refresh the platform's game list.
2. Open **Setup -> Games** on the Controller. Add the exact filename and choose its profile if it is not already listed, then select **Validate** and **Save**. Matching is exact and case-insensitive, and includes the extension.
3. Follow the platform note below.

Platform	After adding the ROM
RetroPie	Copy games into ~/RetroPie/roms/nnes and restart EmulationStation or refresh its game list. Select lr-nestopia-powerglove from the per-ROM launch menu only when you want native Super Glove Ball.
Recalbox	Copy games into /recalbox/share/roms/nnes and update the game list. Restart the VirtualGlove service or reboot after registering Super Glove Ball so its automatic core choice is refreshed.
Batocera	Copy games into /userdata/roms/nnes and update the game list. Restart the VirtualGlove service or reboot after registering Super Glove Ball so its automatic core choice is refreshed.
LaunchBox	Import the game into Nintendo Entertainment System . New games inherit VirtualGlove RetroArch . The launcher checks the registry on every start, so no service restart is needed.

An existing explicit per-game emulator choice is preserved. Registering a game changes only VirtualGlove's profile and launch behavior; it does not copy, rename, or modify the ROM.

4. Pair the devices

Pairing securely gives the Controller and console the same private credential. It does not start controller output.

1. Open the secure Setup address, normally <https://virtualglove.local:8443/setup>. A first visit may show a privacy warning because the Controller is a private local device.
2. Under **Connection and startup**, choose RetroPie, Recalbox, Batocera, or LaunchBox. Enter the console hostname or IP address, then select **Save connection**. Pairing stays unavailable until both fields are saved.
3. Select **Check console address**. This confirms that the saved name or address can be reached; you can run the check again whenever needed.
4. Under **Pair this Controller**, choose **One-time code (recommended)** and select **Continue**.
5. Compare the Matrix ID with the beginning of the browser certificate's SHA-256 fingerprint. If they differ, stop. If they match, confirm the match, enter the six-digit Matrix approval PIN, and continue.
6. Run the one-time pairing command shown by Setup on the selected console. Enter its 20-character code in the browser within five minutes, then select **Pair with console**.
7. Wait for **Pairing complete**. This confirms an authenticated connection; a game test is still required.

Pair with RetroPie

Pair this Controller with your saved RetroPie console. Both methods use a physical confirmation on the Controller matrix.

Saved console: RETROPIE-NAME.local [Change](#)

1. Choose a method 2. Confirm your Controller 3. Pair with RetroPie

1. Choose a pairing method

How will you connect?

☒ One-time code (recommended)
Run a command on RetroPie and enter the code it displays.

☐ SSH password
Use your RetroPie username and password. SSH password login must be enabled.

[Continue](#)

Guided pairing starts with the saved console and a choice of one-time code or SSH password.

Pair with RetroPie

Pair this Controller with your saved RetroPie console. Both methods use a physical confirmation on the Controller matrix.

Saved console: RETROPIE-NAME.local

1. Choose a method

2. Confirm your Controller

3. Pair with RetroPie

RETROPIE-NAME.local · One-time code

Confirmation window: 120 seconds remaining. Console and method are fixed until it ends.

2. Confirm your Controller

The Controller matrix shows ID 1234ABCD, then a six-digit approval PIN. Compare the browser certificate with the physical matrix before entering that PIN.

► How to compare the certificate

☐ I compared the browser certificate fingerprint with the matrix ID and they match

Controller approval PIN

Six digits shown on the Controller matrix

Continue

Controller confirmation with certificate comparison, matrix approval PIN, and remaining time.

Pair with RetroPie

Pair this Controller with your saved RetroPie console. Both methods use a physical confirmation on the Controller matrix.

Saved console: RETROPIE-NAME.local

1. Choose a method

2. Confirm your Controller

3. Pair with RetroPie

RETROPIE-NAME.local · One-time code · Controller confirmation entered

Pairing in progress

Sending the pairing request to RetroPie. Keep this page open while we wait for its response.

SSH pairing can take a few minutes. Pairing does not start controller output.

Pairing in progress. Waiting for RetroPie...

Pairing in progress while the request waits for the selected console.

Pair with RetroPie

Pair this Controller with your saved RetroPie console. Both methods use a physical confirmation on the Controller matrix.

Saved console: RETROPIE-NAME.local [Change](#)

1. Choose a method

2. Confirm your Controller

3. Pair with RetroPie

RETROPIE-NAME.local · One-time code · Controller confirmation entered

Pairing complete

The RetroPie receiver was restarted. The matrix resumes its normal display; when idle, it follows your attract setting. Open Dashboard to start controller output when you are ready. Pairing does not verify that a game received input.

Open Dashboard

Pairing complete. The RetroPie receiver was restarted.



Pairing complete, with the next step on Dashboard.

The confirmation window lasts two minutes. If it expires or pairing fails, select **Start a new confirmation** and obtain a new console code.

Optional: pair with your console password

RetroPie, Recalbox, and Batocera can instead use **SSH password** in the pairing card. Complete the same certificate and Matrix PIN checks, then enter the console's SSH username and password. The Controller uses the password only for that attempt and does not save it.

LaunchBox uses one-time-code pairing only and never requests a Windows password. If neither method works, see [Pairing and token management](#).

Optional: trust the Controller certificate

After confirming that the Matrix ID matches the browser certificate, Setup can download a trust certificate for that phone or computer. Installing it removes future privacy warnings. Never install it if the IDs do not match.

5. Calibrate and test a game

Center your hand

1. Open Dashboard and choose the correct player.
2. Position the camera so your whole hand remains visible throughout the area where you intend to move.
3. Hold a relaxed open hand at a comfortable center and distance.
4. Select **Center hand** and remain still until it reports completion.

Center again after moving the camera, changing playing position, or switching to a player who has not been calibrated. The saved center belongs to the player and does not chase the hand during play.

For a guided first session, choose **Get ready to play** on Setup or Dashboard. It checks the connection, camera, center, and essential gestures without sending game controls until you explicitly finish practice.

Test an ordinary game

1. On Dashboard, select **Start controller**.
2. Launch a registered NES game that uses FCEUmm.
3. Confirm the expected profile appears on Dashboard and the Matrix.
4. Test Left, Right, Up, Down, A, B, Start, and Select.
5. Confirm the physical controller still works for Player 1.
6. Return the hand to center and confirm movement stops.
7. Exit the game and confirm gesture output stops.

Different Programs deliberately use different gestures. If a control does not behave as expected, check the selected Program in the [Gameplay Guide](#) before changing calibration.

Checkpoint: A gesture must change the intended control inside the running game. A running service or visible controller name alone is not an end-to-end test.

Test Super Glove Ball

- With FCEUmm, test the ordinary joystick profile first.
- With Nestopia (VirtualGlove), test Start, hand movement, forward and backward depth, grab and throw, index-point fire, and Power Punch.
- On RetroPie, choose the native core from the ROM's launch menu. Recalbox, Batocera, and LaunchBox select it automatically for an exact registered ROM unless you have made another explicit choice.

If the native core is unavailable or unsuitable, return that ROM to FCEUmm; the rest of VirtualGlove remains installed.

6. Confirm startup and finish

1. Reboot the Controller and console normally.
2. Confirm that Dashboard returns and the console reconnects without pairing again.
3. Confirm the selected player, center, tuning, camera choices, and game registrations remain available.
4. Launch the ordinary FCEUmm test game again and confirm both VirtualGlove and the physical controller work.
5. If installed, test Super Glove Ball again.

The Controller remembers whether you selected **Start controller** or **Stop controller**, but controls are delivered only during a recognized game session or an intentional manual profile. Unknown games and game exit release all controls safely.

Updates and checks

To update, close running games and repeat the same install command on the Controller and console. Use the same release on both devices. The installers stop managed VirtualGlove processes before replacing application files and preserve private settings and user data.

Upgrade from v0.4.2 to v0.5.0

Version 0.5.0 includes a managed upgrade from the released v0.4.2 installation. Update the Controller and console from the same v0.5.0 release. The installers back up and remove retired application files while preserving:

- players, calibration, tuning, and dead-zone settings;
- camera and device configuration;
- pairing and the saved console address;
- game registrations and rapid-fire choices;
- ROMs, saves, controller assignments, and installed native cores.

Keep the printed backup location until you have completed the reboot and game checks above. If the installer reports an unknown or locally modified retired file, follow its message rather than deleting the installation manifest or forcing the upgrade. The [technical installation reference](#) describes staged upgrades and recovery.

Read the installer report

Result	What it means
PASS	That check succeeded.
ACTION	Installation succeeded, but you still need to complete the named step, such as pairing or testing a game.
FAIL	Resolve the reported problem before continuing, then run the installer again.

A technical **PASS** cannot prove that your camera sees your hand or that a game responded. Complete the physical game checks before considering installation finished.

If something does not work

The download fails

Confirm internet access and retry the same command. A failed download or checksum installs nothing. If the latest stable release does not contain the required installer asset, wait for the completed release rather than using a file from a different version.

Dashboard or Setup does not open

Try the Controller's IP address instead of its `.local` name:

- Dashboard: <http://CONTROLLER-IP:8088/dashboard>
- secure Setup: <https://CONTROLLER-IP:8443/setup>

Confirm the Controller and browser are on the same local network. The Matrix hourglass means VirtualGlove is still starting; a blinking X means Dashboard has an error to report.

The camera is unavailable

Open **Play** or **Glove Academy** and wait for the camera view. Reconnect the camera, powered hub, and cable if it remains unavailable. The recovery helper enrolls a supported camera after its first successful frame, even when no camera was connected during installation. Do not change advanced camera settings until the automatic configuration has been tested.

Pairing fails or expires

Confirm the saved platform and address, run **Check console address**, and start a new Controller confirmation. Obtain a new console code; expired PINs and one-time codes cannot be reused. LaunchBox does not support SSH-password pairing. Never copy pairing credentials between consoles manually.

The game launches but hand controls do not work

Check these in order:

1. Dashboard shows **Start controller** as active.
2. The game filename is registered under **Setup -> Games**.

3. Dashboard shows the expected active game and profile.
4. The camera sees a calibrated hand.
5. The platform uses the VirtualGlove controller arrangement described in its install section.

Stop and start the Controller once, then relaunch the game. Do not replace the entire RetroArch configuration to repair one binding.

The wrong physical controller is Player 1

On Recalbox or Batocera, close the game and rerun the installer. Choose the correct physical Player 1 controller when prompted. On RetroPie or LaunchBox, use the platform's normal controller assignment while keeping VirtualGlove's managed entry intact.

Super Glove Ball uses the wrong core

Confirm that the exact ROM filename is registered for Super Glove Ball. On RetroPie, choose [lr-nestopia-powerglove](#) from the per-ROM launch menu. On Recalbox, Batocera, or LaunchBox, restart the platform integration after adding the ROM and confirm that no explicit per-game emulator choice overrides the automatic selection. FCEUmm remains a safe fallback.

Installation stops partway through

Correct the reported problem and rerun the same installer. Keep the backup path printed by the installer. For service commands, logs, manual repairs, or package details, use the [Troubleshooting Guide](#) and [installation troubleshooting commands](#).

Matrix and web-page quick reference

These are the Matrix states most useful during installation:

Display	Meaning
System heart or pulsing hourglass	The Controller is starting.
Animated glove	Gestures are off and the Controller is idle.
Scanning L	Play or Glove Academy practice is active; game output is paused.
1-14 , A-I , BS , or GB	The corresponding game profile is selected.
Pulsing profile code	A calibrated hand is being tracked.
Blinking X	Open Dashboard to read the reported problem.
Blank	Check Controller power and Dashboard; blank does not confirm shutdown.

See the [Matrix Guide](#) for every animation and display state.

Page	Normal address
Dashboard	http://UNO-Q-NAME.local:8088/dashboard

Page	Normal address
Play	http://UNO-Q-NAME.local:8088/play
Glove Academy	http://UNO-Q-NAME.local:8088/learn
Setup and Games	http://UNO-Q-NAME.local:8088/setup
Secure pairing	https://UNO-Q-NAME.local:8443/setup
Help and printable manuals	http://UNO-Q-NAME.local:8088/help

For gestures and game controls, use the [Gameplay Guide](#). Technical details are in [Architecture](#), [VirtualGlove Input Modes](#), and the [Configuration Reference](#).